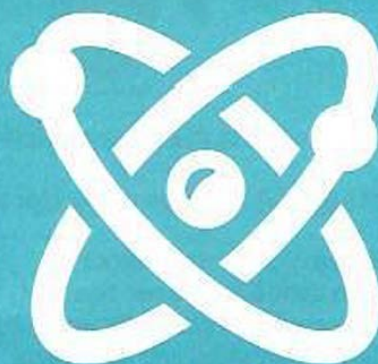


ICAS

PAPER

C



2017 SCIENCE

**DO NOT OPEN THIS BOOKLET
UNTIL INSTRUCTED.**

40 QUESTIONS

TIME ALLOWED: 55 MINUTES

Read the instructions on the **ANSWER SHEET** and fill in your **NAME SCHOOL** and **OTHER INFORMATION**.

Use a pencil. Do **NOT** use a coloured pencil or a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. Sue wants to make coloured eggs by dyeing the eggshells with plant juice.

The diagram shows the eggshell colours that some plant juices produce.



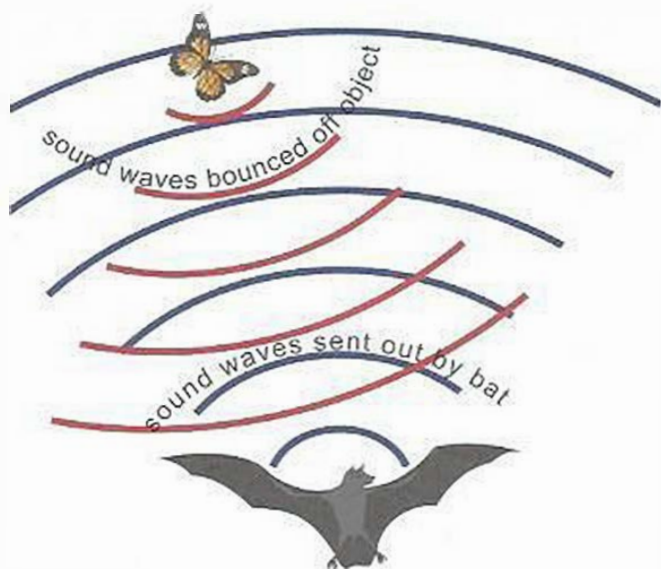
Mixing blue dye and red dye makes purple dye.

Which plant juices should Sue mix to dye the eggshells purple?

- (A) elderberry and red beetroot
- (B) purple cabbage and red beetroot
- (C) purple cabbage and elderberry
- (D) yellow onion skins and elderberry

2. Echoes are produced when sound waves bounce off solid objects. Bats use echoes to determine the distance between them and objects in their surroundings.

A bat sends out a series of high-pitched sounds. The times it took for sound waves to reach some objects and bounce back to the bat are recorded in the table.



Object	Time taken for sound to reach object and bounce back to bat (seconds)
butterfly	0.25
tree	0.30
building	0.35
bird	0.20

Based on the data in the table, which conclusion is correct?

- (A) The bird is flying towards the bat.
- (B) The building is the furthest object from the bat.
- (C) The butterfly is the closest object to the bat.
- (D) The bird is further from the bat than the building.

3. Like humans, chimpanzees make sounds to express emotions.

The table lists some sounds and the emotion each expresses.

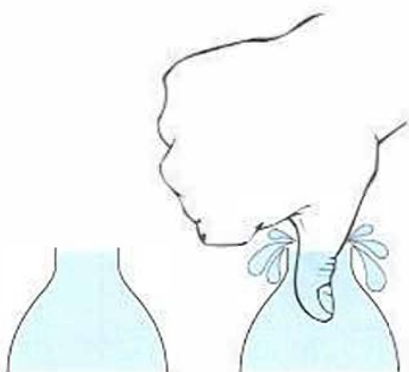
Sound	Emotion
'wraa'	fear
pant-hoot	excitement
lip smack, pant	happiness
soft bark, cough	annoyance
scream, bark	fear, anger
'hoo', whimper	distress

One chimpanzee makes pant-hoot and lip smack noises when seeing another chimpanzee.

What emotions is the chimpanzee expressing?

- (A) fear and distress
- (B) excitement and distress
- (C) distress and annoyance
- (D) excitement and happiness

4. In an experiment, the volume of a thumb was measured as shown in the diagram.



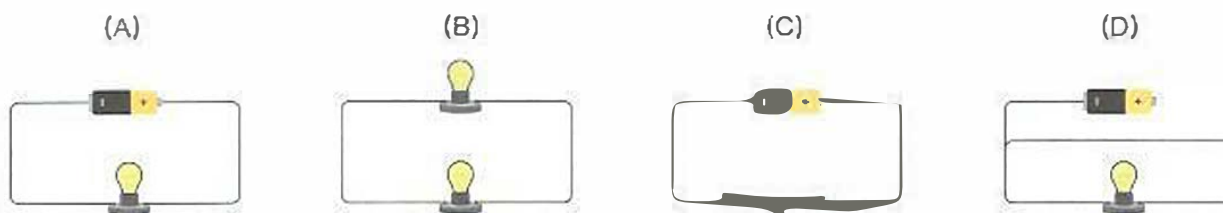
Which of the following methods will give the volume of the thumb?

- (A) measure how much water is in the bottle
- (B) measure how much water is left on the thumb
- (C) measure how much water it takes to refill the bottle
- (D) measure how deep the thumb goes into the water

5. For an electric circuit to work, the circuit requires:

- a power source such as a battery,
- an unbroken pathway of metal wire,
- components, such as bulbs or motors, that allow electricity to pass through them.

Which circuit will have electricity flowing around it?



6. Kyle measured a plant using a ruler, as shown.



How many centimetres long is the stem of the plant?

- (A) $4\frac{1}{2}$
 (B) 5
 (C) $5\frac{1}{2}$
 (D) 7

7. Each planet in the Solar System spins on its axis while it moves around the Sun, just like Earth. The time it takes a planet to spin one full rotation on its axis is called a day.

The table lists the approximate duration of a day on different planets in the Solar System.



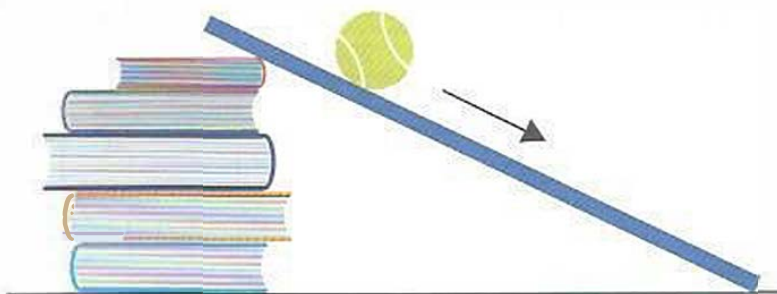
Planet	Duration of day (Earth hours)
Mercury	1408
Venus	5832
Earth	24
Mars	25
Jupiter	10
Saturn	11
Uranus	17
Neptune	16

How many planets have a day that is longer than a day on Earth?

- (A) 4 (B) 3 (C) 2 (D) 1

8. Dylan wants to investigate whether changing the slope of a ramp affects how long it takes a ball to roll down it.

He set up a ramp as shown.



Dylan can select only one extra piece of equipment to carry out this investigation.

Which of the following should he select?

- | | |
|-----------------|--------------------|
| (A) a stopwatch | (B) a thicker book |
| (C) a ruler | (D) a heavier ball |

9. A car travelling at 60 km per hour leaked oil onto the road. An oil drop fell onto the road every second, making this pattern on the road:



Later, the car travelled at 90 km per hour. What did the oil drop pattern on the road look like at this speed?

- (A)
- (B)
- (C)
- (D)

10. Veronica measured the temperature in her school playground three times a day over four days. The table shows her results.

Day	Highest temperature (°C)	Lowest temperature (°C)	Temperature at 11am (°C)
1	13	2	7
2	15	3	6
3	11	14	6
4	10	5	7

Veronica realised that she had made an error in recording the data.

On what day did Veronica make the error?

- (A) Day 1
- (B) Day 2
- (C) Day 3
- (D) Day 4

11. This is a photograph of the sole of one of Bernie's shoes.



Which of these is a photograph of the other shoe from this pair?

(A)



(B)



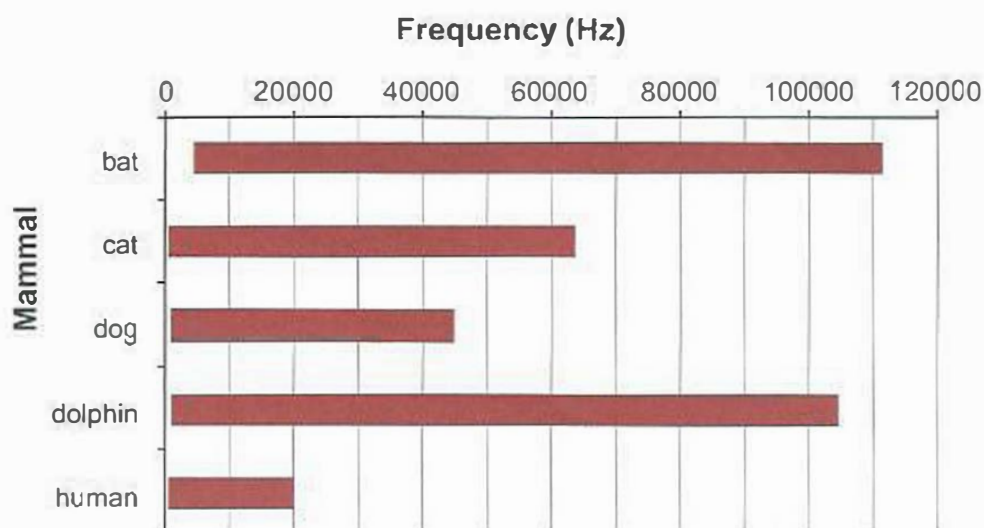
(C)



(D)



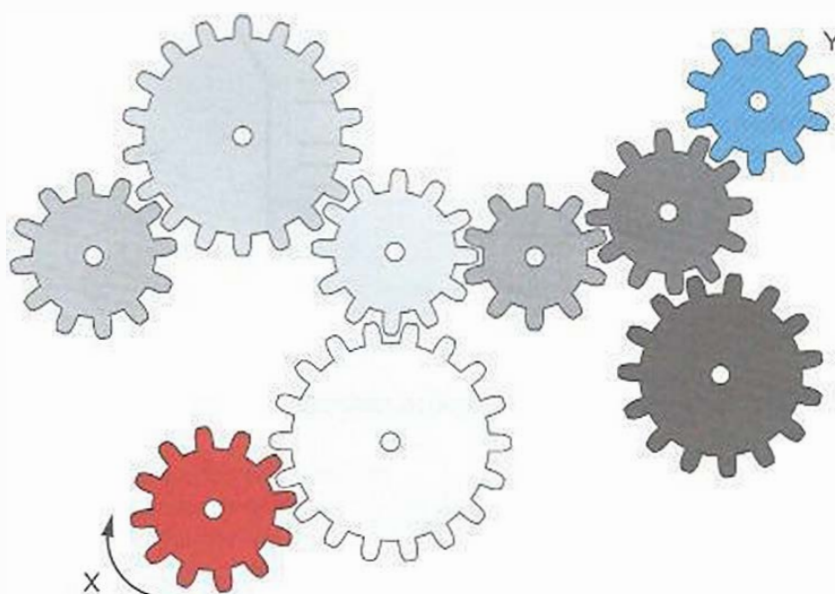
12. The frequency of a sound, measured in hertz (Hz), is what gives the sound its pitch. Different animals can hear sounds in different frequency ranges. This graph shows the hearing range of some mammals.



Which of these mammals can hear a sound with a frequency of 50000 Hz?

- (A) bat and dolphin only
(B) bat, cat and dolphin
(C) bat, cat and dog
(D) bat, cat, dog and dolphin
13. When one gear turns another gear, the gear with more teeth will rotate more slowly than the gear with fewer teeth. The two gears will turn in opposite directions.

The diagram represents a number of gears joined together.



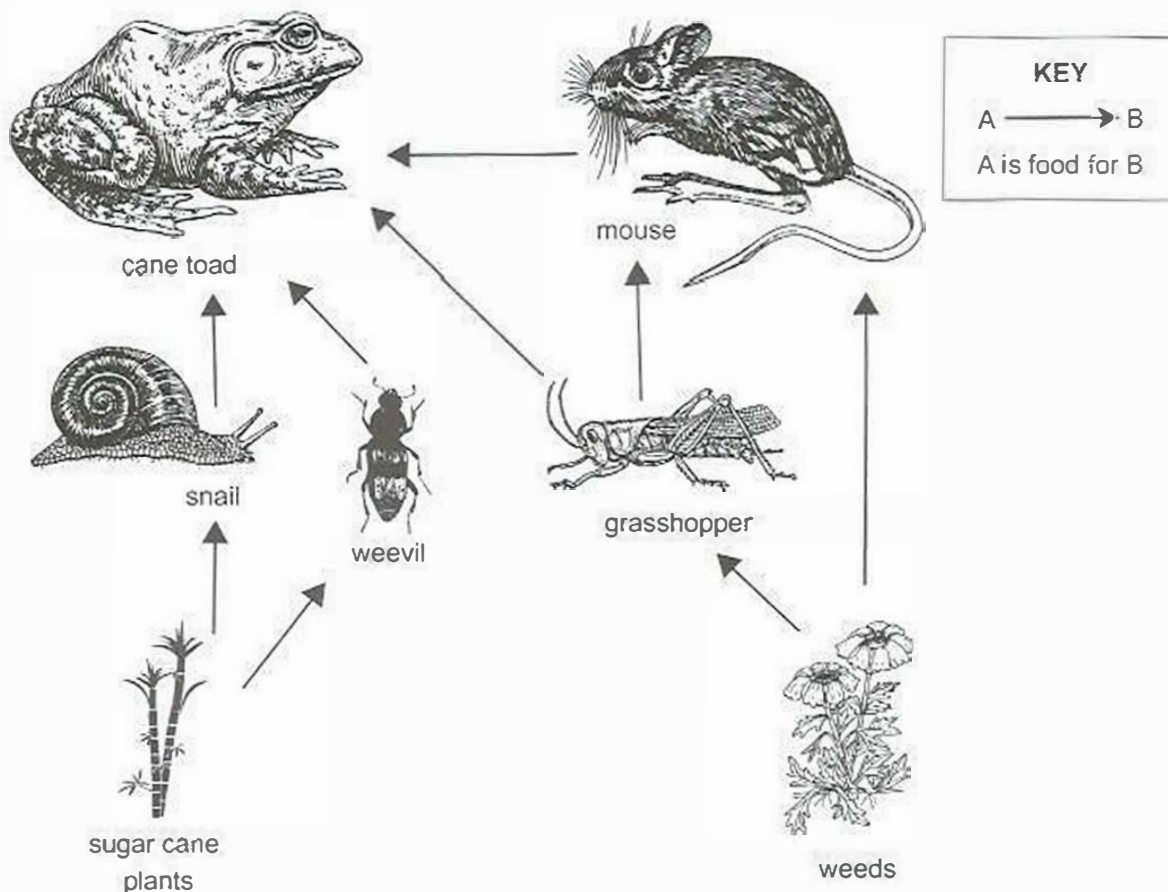
Gear X turns clockwise.

Which option describes the turning of gear Y?

- (A) clockwise and faster than gear X
(B) clockwise and slower than gear X
(C) anticlockwise and faster than gear X
(D) anticlockwise and slower than gear X

14. Prey is an animal that gets eaten by another animal, a predator.

The diagram represents the relationship between plants and animals living on a sugar cane farm in Australia.



Which of the following are predators of grasshoppers?

- (A) weeds and mouse
- (B) mouse and weevil
- (C) mouse and cane toad
- (D) cane toad and weevil

15. A key allows users to identify an object, such as an animal or a plant, using the characteristics of the object.

The diagram shows a key for a group of animals called vertebrates.

1.	has feathers	bird
	does not have feathers	go to 2
2.	has scales	go to 3
	does not have scales	go to 4
3.	has gills	fish
	does not have gills	reptile
4.	has fur	mammal
	does not have fur	amphibian

A caecilian does not have feathers, does not have scales and does not have fur.

What type of vertebrate is a caecilian?

- (A) fish
- (B) reptile
- (C) mammal
- (D) amphibian

16. Igneous rock and sedimentary rock are two types of rocks. Igneous rocks form when hot liquid rock below Earth's surface cools and hardens. Sedimentary rocks form when existing rock pieces and sediment are pressed together.

Alex used chocolate to demonstrate the formation of rocks of different types, using these steps:

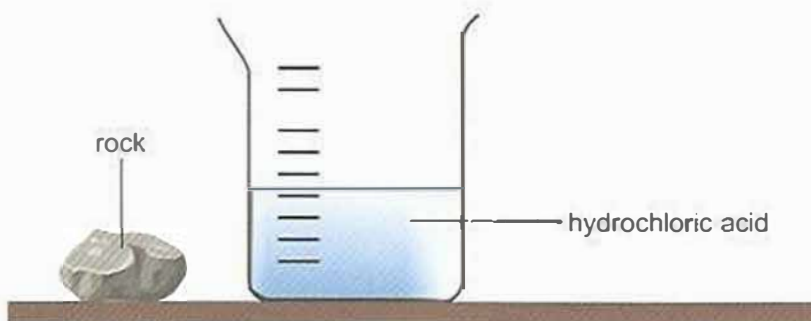
1. Place a container with chocolate in a tub of hot water until the chocolate melts.
2. Take the container out of the hot water and let the chocolate cool.
3. Once the chocolate is hard again, scrape it into small pieces.
4. Press the pieces down to make a block.



Which option correctly matches the steps in Alex's demonstration with the formation of igneous rocks and sedimentary rocks?

	Formation of igneous rocks	Formation of sedimentary rocks
(A)	Step 2	Step 4
(B)	Step 4	Step 2
(C)	Step 1	Step 3
(D)	Step 2	Step 3

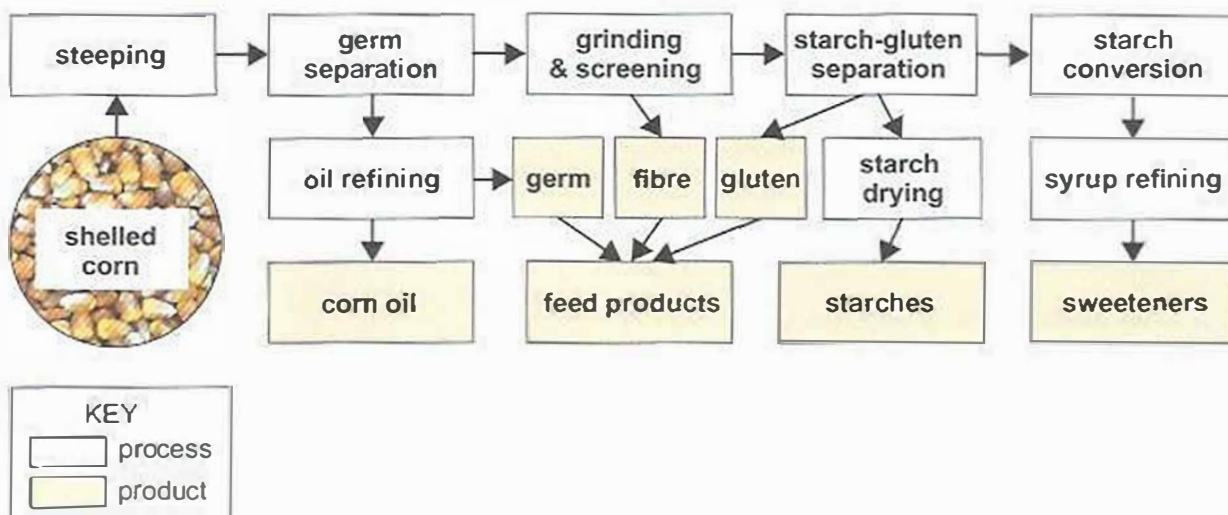
17. Kate's teacher gave Kate some hydrochloric acid and a rock. She told Kate that some rocks will react with hydrochloric acid to produce carbon dioxide. Carbon dioxide is a gas that has no colour or smell.



How can Kate find out whether this rock produces carbon dioxide?

- put the rock in the container and smell it
- put the rock in the container and look for bubbles in the liquid
- put the rock in the container and look for a change of colour
- put the rock in the container and look for a rise in water level

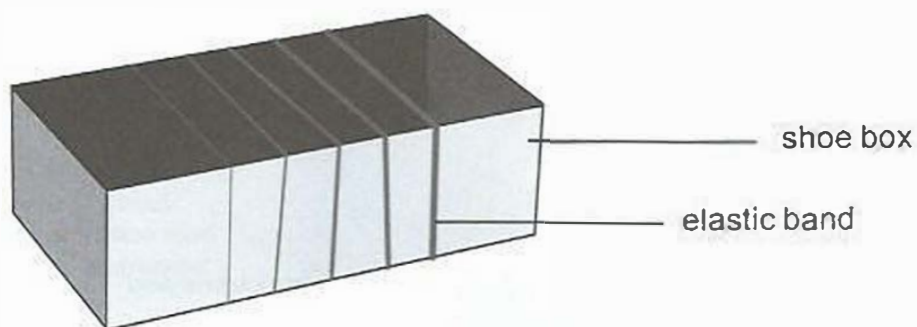
18. The diagram summarises the steps in producing a number of products from corn.



Which of these steps does NOT produce feed products?

- (A) oil refining
- (B) grinding and screening
- (C) starch-gluten separation
- (D) starch drying

19. Megan made a musical instrument using some elastic bands and an empty shoe box. The elastic bands all had different lengths and thicknesses. When she plucked them, each band made a sound with a different pitch.

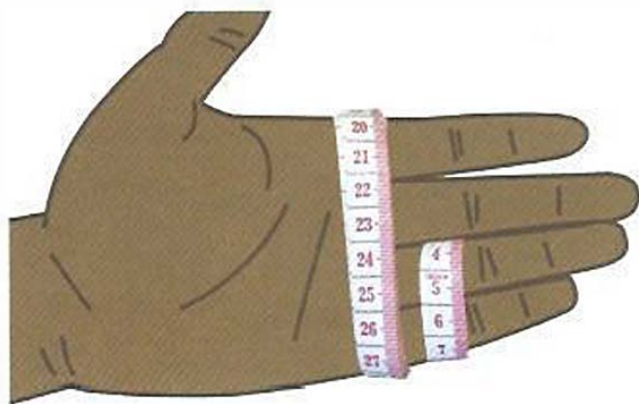


Megan was investigating whether the thickness of an elastic band affects the pitch of the sound it produces when plucked.

What must she do to make her investigation a fair test?

- (A) pluck each elastic band three times
- (B) use elastic bands made from different types of rubber
- (C) use elastic bands with exactly the same thickness but different lengths
- (D) use elastic bands with exactly the same length but different thicknesses

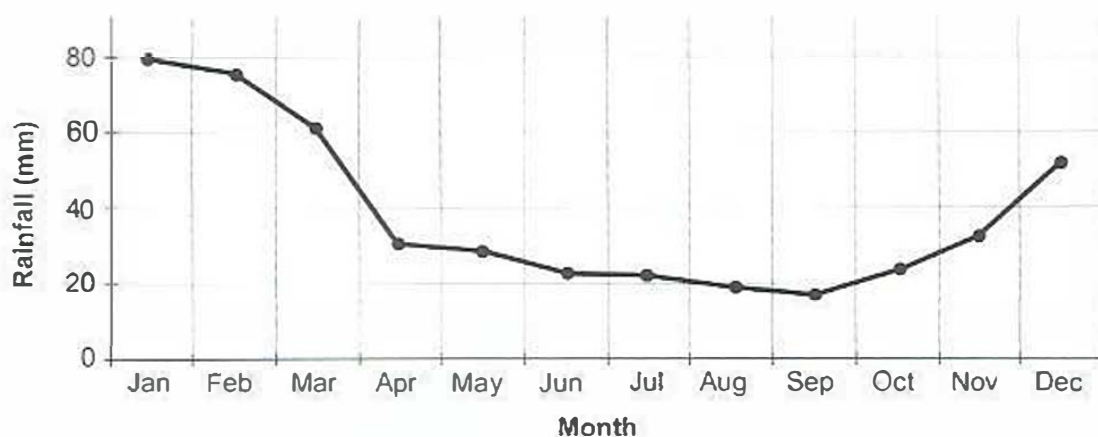
20. Jack wrapped a measuring tape around his fingers several times, as shown.



Approximately how many centimetres wide is each of Jack's fingers at the base?

- (A) 3.5
- (B) 2
- (C) 1.5
- (D) 1

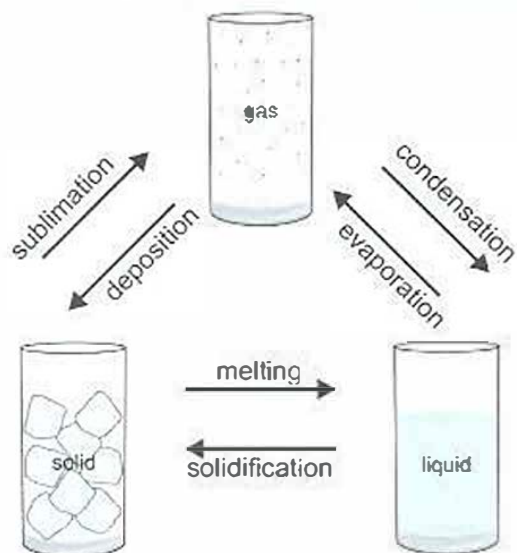
21. The graph summarises the average monthly rainfall for the years 1961–1990.



Which of the following periods had the lowest average amount of rainfall?

- (A) Jan – Mar
- (B) Apr – Jun
- (C) July – Sep
- (D) Oct – Dec

22. Matter can change from one state to another by heating or cooling. The diagram shows the changes of state for a substance.



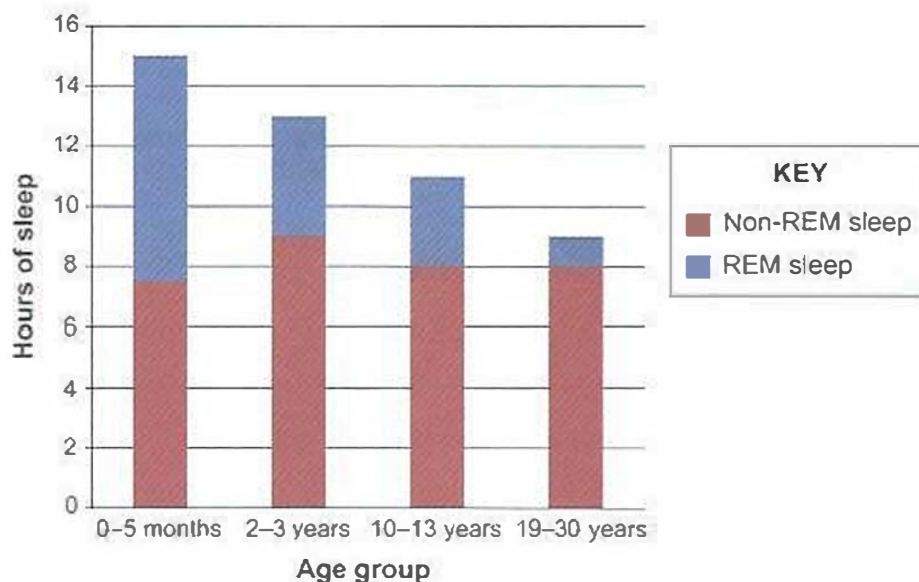
Iodine is a purple solid at room temperature. When placed in a warm container, it fills the container with a purple gas without melting.

Which process does iodine undergo when heated?

- (A) sublimation
- (B) evaporation
- (C) deposition
- (D) solidification

23. Rapid eye movement (REM) sleep is a stage during sleep when the eyes of the sleeping person move rapidly.

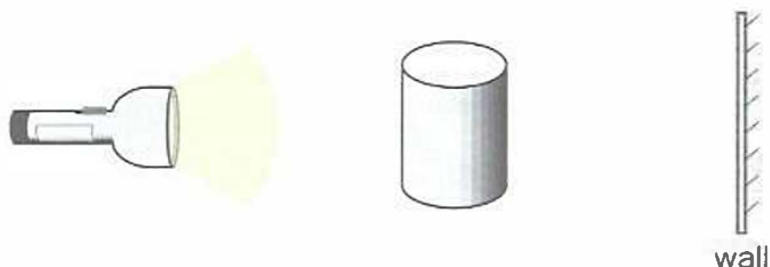
The graph represents the hours of REM and non-REM sleep for people in different age groups.



How many hours more than the 19-30 year age group does the 0-5 month age group spend sleeping per day?

- (A) 15 (B) 9 (C) 7 (D) 6

24. Joyce shines a torch onto a cylinder, as shown.



Which diagram shows the shadow produced by the cylinder on the wall?



25. The table classifies flowers by their heights when fully grown.

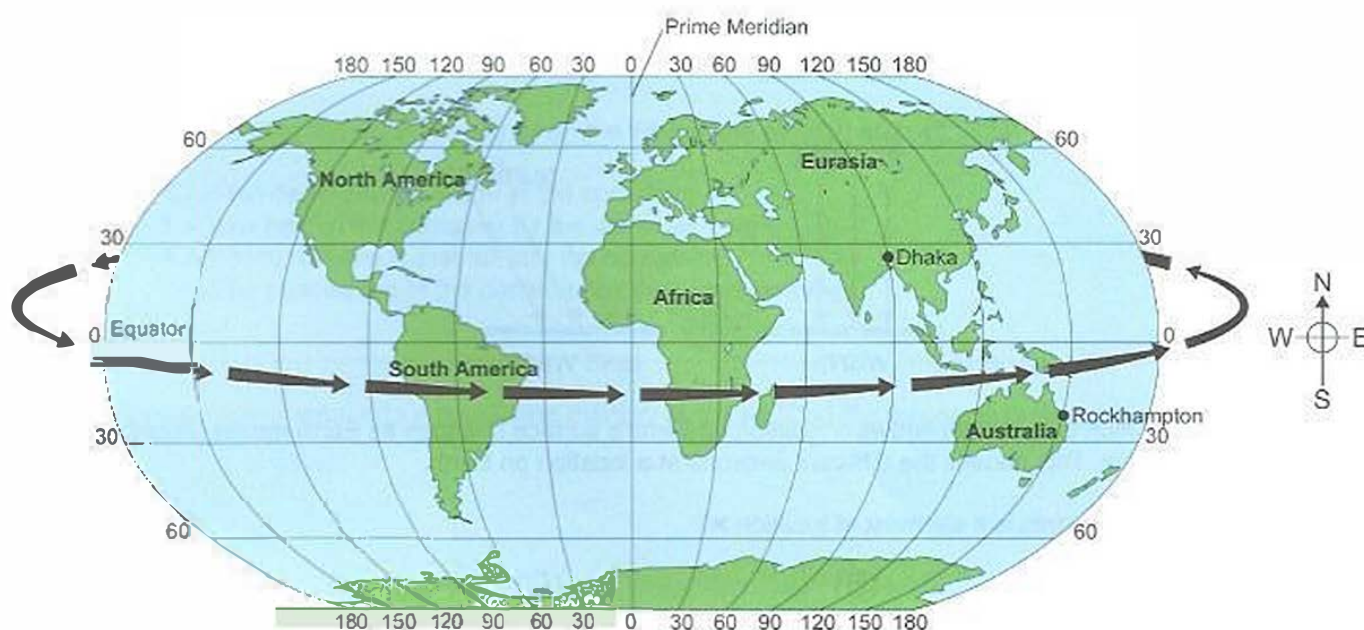
Short		Medium		Tall	
Name	Height (cm)	Name	Height (cm)	Name	Height (cm)
Grecian windflower	5 – 15	Lily of the valley	15 – 30	Spanish bluebell	25 – 46
Siberian squill	12 – 20	Jonquilla daffodil	25 – 40	Lily-flowered tulip	36 – 60
Dwarf iris	10 – 15	Hyacinth	20 – 30	Short cup daffodil	30 – 46

Which statement is correct based on the table?

- (A) A flower with a height of 34 cm when fully grown must be a tall flower.
 (B) A flower with a height of 15 cm when fully grown must be a short flower.
 (C) A flower with a height of 20 cm when fully grown must be a short flower.
 (D) A flower with a height of 23 cm when fully grown must be a medium flower.

26. Places on the surface of Earth are given a location using coordinates called latitude and longitude. These coordinates locate places by their distance from the Equator and the Prime Meridian.

One complete revolution of Earth occurs once every 24 hours. The planet is divided into 24 time zones one hour apart.



A fossil was found at location 20° N and 75° E.

Where was the fossil found?

- (A) Eurasia
 (B) Australia
 (C) Africa
 (D) South America

27. The chemical formula for water is H_2O and can be represented by the following diagram:

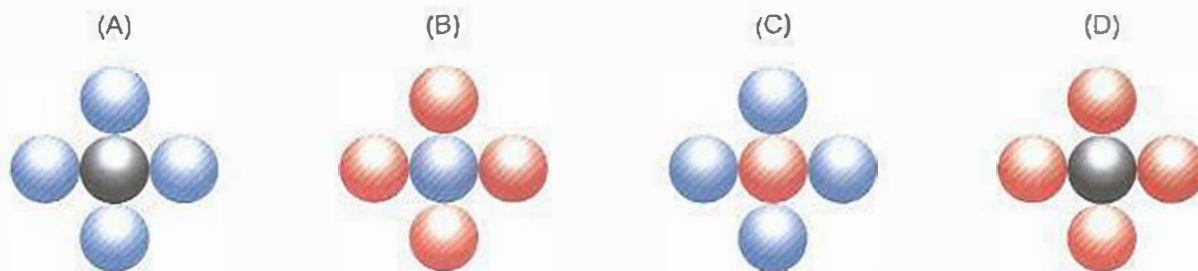


The chemical formula for carbon dioxide is CO_2 and can be represented by the following diagram:

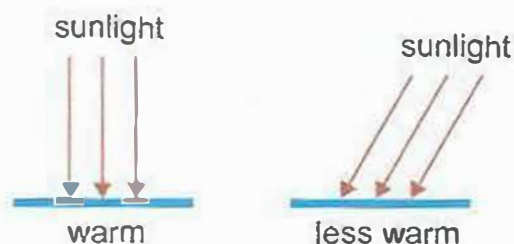


Methane, the main gas found in natural gas, has the chemical formula CH_4 .

Which diagram could represent methane?

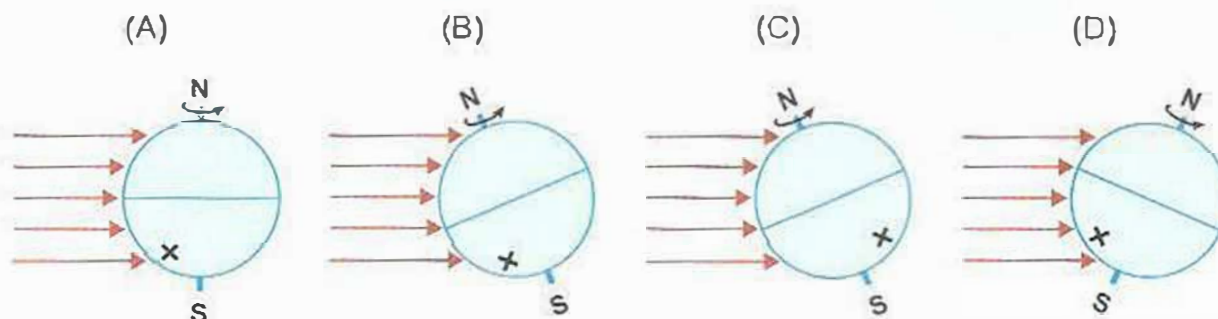


28. When a surface receives sunlight, the angle at which the sunlight strikes the surface affects the amount of heat the surface receives.



The angle at which the sunlight strikes a location on Earth's surface changes as Earth moves around the Sun during a year. This causes the different seasons at a location on Earth.

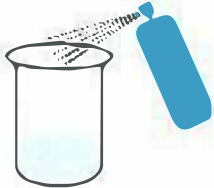
In which of the diagrams is it warmest at location x?



For questions 29 and 30 use the information below.

Warm air is less dense than cold air and tends to rise. When warm air containing water vapour rises and cools down, the water vapour condenses onto small particles floating in the air. Eventually a cloud forms.

Jo carried out this experiment to demonstrate the formation of a cloud.

Step 1	Step 2	Step 3	Step 4
			
Pour some boiled water into a container and swirl it to heat up the sides.	Place ice cubes on the lid of the container for a few minutes.	Take the lid off and quickly squirt in some hairspray.	Put the lid with the ice on it, back on top of the container. Watch the cloud form.

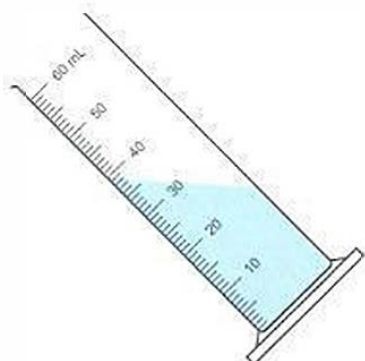
29. What is the purpose of using the hairspray?

- | | |
|----------------------------------|----------------------------|
| (A) to add warm vapour | (B) to cool down the air |
| (C) to introduce small particles | (D) to make the cloud rain |

30. What will happen to the cloud formed when the lid is taken off after step 4?

- (A) It will fall down into the water in the container.
(B) It will be held in the container by the warm air inside.
(C) It will form rain drops that fall into the container.
(D) It will be pushed out of the container by the warm air inside.

31. Tara poured some water into a measuring cylinder and then tilted the measuring cylinder as shown.



How much water (mL) is in the measuring cylinder?

- (A) 20
(B) 30
(C) 35
(D) 40

32. One serving of food is the amount of a type of food that an average person would normally eat in one meal. By measuring food in servings, it is possible to compare the nutrients and energy content (kJ) in foods.

The table outlines the energy content in one serving of some different foods.

Food	Serving size	Energy (kJ)
Soft drink	200 mL	21
Popcorn	1 large cup	65
Pepperoni pizza	1 slice	120
Cheeseburger	1 burger	80

According to the table, which of these has the greatest energy content?

- (A) 600 mL soft drink
- (B) 1 slice of pepperoni pizza
- (C) 2 large cups of popcorn
- (D) 2 cheeseburgers

For questions 33 and 34 use the information below.

Plants need light, water and nutrients to grow. Plants normally get their nutrients from the soil.

Ben wanted to investigate what is needed to sprout and grow cress seeds. He put some cress seeds on paper towels in a bowl near the window. For ten days he added a small amount of water each day to keep the paper towel wet.

Ben summarised his observations in these three drawings:



33. What conclusion could Ben draw?

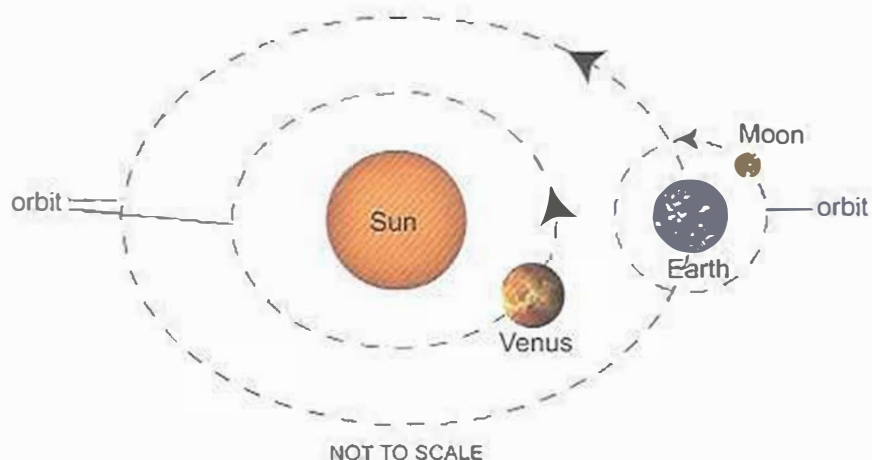
- (A) Cress can only live for 5 to 10 days.
- (B) Cress seeds need water to sprout but do not need water to grow.
- (C) Cress seeds need more than just water to keep alive.
- (D) Cress seeds use water and the nutrients in the paper towels to grow.

34. What does Ben need to do to find out whether light affects the sprouting?

- (A) Repeat exactly what he did in the first investigation.
- (B) Repeat the investigation with the bowl in a cupboard.
- (C) Repeat the investigation with the bowl in a cupboard with no water to be added.
- (D) Repeat the investigation with the bowl placed outside the window.

35. A satellite is any object that moves around a larger object in space. The path that a satellite follows, as it moves around the larger object, is called its orbit.

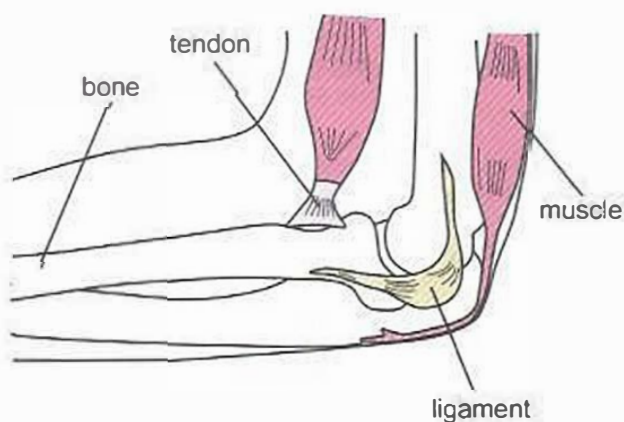
The diagram shows the movement of Venus, Earth and the Moon in relation to the Sun.



Which statement is correct?

- (A) Earth is in orbit around the Moon and Venus is a satellite of the Sun.
 (B) The Moon is in orbit around Earth and Earth is a satellite of Venus.
 (C) Earth is a satellite of the Moon and the Moon is in orbit around Venus.
 (D) The Moon is a satellite of Earth and Venus is in orbit around the Sun.
36. Our skeleton consists of bones that give our body shape and allow us to move. The bones are moved by muscles. Tendons and ligaments are connective tissues.

The diagram represents an elbow joint.



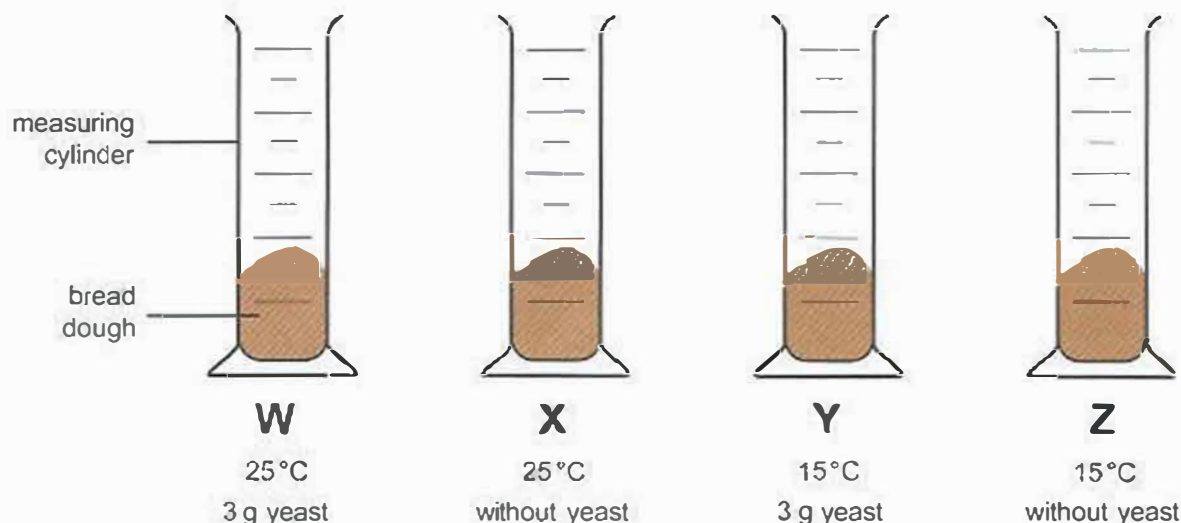
Which of these identifies what ligaments and tendons connect?

	Ligaments connect	Tendons connect
(A)	bone and muscle.	bone and bone.
(B)	bone and bone.	bone and muscle.
(C)	muscle and muscle.	bone and bone.
(D)	bone and bone.	muscle and muscle.

For questions 37 and 38 use the information below.

Michelle read that bread dough needs yeast and warmth to rise.

She set up an experiment to investigate this, as shown.



She left each measuring cylinder for 24 hours and recorded how high each piece of dough rose.

37. To find out the effect of the presence of yeast on how much the dough rises, which two cylinders can Michelle compare?

- (A) W and X
- (B) X and Y
- (C) W and Y
- (D) W and Z

38. Michelle read that the more yeast added, the higher the dough will rise; the higher the temperature, the higher the dough will rise.

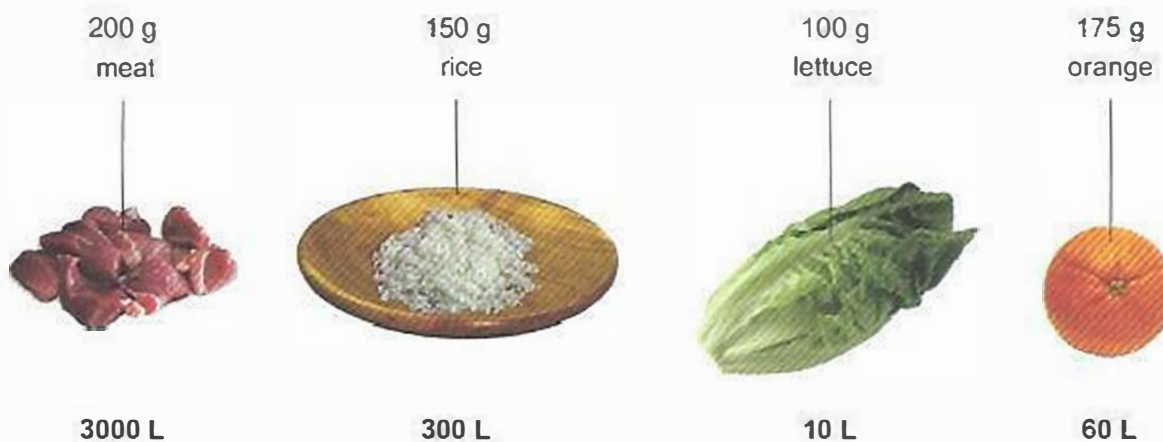
Michelle then set up another cylinder, V, with the same amount of dough and 6 g of yeast added. She kept it at 20 °C for 24 hours, and recorded the height of the dough.

Which statement **MUST** be true based on what Michelle read?

- (A) The dough in V will rise the highest among all cylinders.
- (B) The dough in W and V will rise the same amount
- (C) The dough in X will rise higher than that in V.
- (D) The dough in V will rise higher than that in Y.

39. Water is needed to produce food.

The diagram shows how much water is needed in producing various amounts of different types of food.

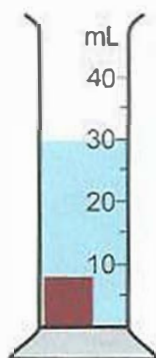


Which statement is supported by the information in the diagram?

- (A) There is less water in an orange than there is in rice.
- (B) It takes less water to grow lettuce than to grow the same mass of rice.
- (C) It takes more water to produce meat because the animals eat lots of plants.
- (D) The more water it takes to produce an amount of food, the heavier this amount of food will be.

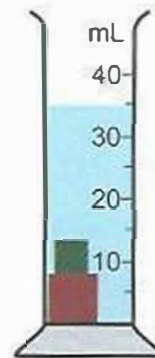
40. The volume of an object is equal to the volume of water it displaces when the object is fully underwater.

Frank placed a block in a measuring cylinder that contained some water, as shown.



He then added another, smaller block.

The volume of the larger block is twice the volume of the smaller one.



How much water is in the measuring cylinder?

- | | |
|-----------|-----------|
| (A) 10 mL | (B) 20 mL |
| (C) 25 mL | (D) 30 mL |

ACKNOWLEDGMENT

Copyright in this booklet is owned by UNSW Global Pty Limited, unless otherwise indicated. Every effort has been made to trace and acknowledge copyright. UNSW Global Pty Limited apologises for any accidental infringement and welcomes information to redress the situation.

SOURCES

Question 39 Lettuce image adapted from an image by NY <<http://www.picserver.org/1Aetluce-2.html>> licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported licence <<https://creativecommons.org/licenses/by-sa/3.0/>>.

THE FOLLOWING YEAR LEVELS SHOULD SIT THIS PAPER:

Australia	Year 5
Brunei	Primary 5
Hong Kong	Primary 5
Indian Subcontinent ¹	Class 5
Indonesia	Year 6
Malaysia	Standard 5
Middle East ²	Class 5
New Zealand/Pacific ³	Year 6
Singapore	Primary 4
Southern Africa ⁴	Grade 5

¹ Indian Subcontinent Region: India, Sri Lanka, Nepal, Bhutan and Bangladesh

² Middle East Region: United Arab Emirates, Qatar, Kuwait, Saudi Arabia, Egypt, Bahrain, Oman, Turkey, Lebanon, Tunisia, Morocco, Libya, Algeria and Jordan

³ Pacific Region: Vanuatu, Papua New Guinea and Fiji

⁴ Southern African Region: South Africa, Botswana, Lesotho, Swaziland, Zimbabwe and Namibia

PAPER

C



© 2017 UNSW Global Pty Limited. Copyright in this document is owned by UNSW Global Pty Limited.

IF THIS COVER APPEARS IN BLACK AND GREY ONLY, THIS PAPER IS AN ILLEGAL COPY.



UNSW Global
AUSTRALIA

ICAS

2017 Science - Year 5

The table below shows all questions. The questions answered incorrectly by XXXXX are shaded. Questions can be sorted by clicking on the table header. Click on the question to view it.

Question	Area	Question	Correct	XXXXX's	AU
1	Interpreting	Interpret information to produce a solution to a problem	C	☒	87
2	Predicting/Concluding	Infer the distance to an object using echo timing	B	☒	74
3	Interpreting	Interpret tabulated information to describe animal behaviour	D	☒	96
4	Observing/Measuring	Measure the volume of an object using displacement	C	☒	50
5	Predicting/Concluding	Use information provided to identify the electric circuit that will work	A	☒	87
6	Observing/Measuring	Determine a length using a ruler	A	☒	69
7	Predicting/Concluding	Interpret information provided in a table	B	☒	88
8	Investigating	Identify the piece of equipment needed to carry out a particular experiment	A	☒	64
9	Reasoning/Problem sol	Predict an oil drop pattern based on the speed of a car	C	A	57
10	Interpreting	Identify the incorrect data that had been entered in a table	C	☒	92
11	Observing/Measuring	Identify a pattern that is the mirror image of another	A	☒	73
12	Observing/Measuring	Interpret information presented in a column graph	B	☒	68
13	Reasoning/Problem sol	Determine the direction and relative speed of a gear	C	☒	58
14	Interpreting	Identify a relationship in a food web	C	☒	90
15	Predicting/Concluding	Use a key to classify a caecilian	D	☒	80
16	Predicting/Concluding	Match steps in a demonstration with steps in a description of a natural process	A	☒	62
17	Investigating	Identify the best way to carry out an investigation	B	☒	66
18	Interpreting	Identify the relationship between elements of a flowchart	D	☒	65
19	Investigating	Identify factors that make a test fair	D	☒	72
20	Observing/Measuring	Determine the average width of a number of objects	B	☒	61
21	Interpreting	Identify a specified value range in a line graph	C	☒	93
22	Interpreting	Use information provided to identify a described physical change	A	☒	56
23	Interpreting	Interpret sector bar graphs	D	☒	68
24	Reasoning/Problem sol	Predict the shape of the shadow that a given object will cast	C	D	48
25	Predicting/Concluding	Apply information in a table to a range of living things	D	B	32
26	Predicting/Concluding	Locate a fossil using map coordinates	A	C	56
27	Reasoning/Problem sol	Match a diagram to its chemical formula	A	B	44
28	Predicting/Concluding	Identify the diagram that correctly represents a given situation	D	☒	60
29	Investigating	Identify the purpose of a step in an investigation	C	☒	52
30	Reasoning/Problem sol	Predict an outcome based on information provided	D	☒	54
31	Predicting/Concluding	Predict the volume of water in a measuring cylinder	B	☒	69
32	Reasoning/Problem sol	Draw a conclusion based on information in a table	D	☒	59
33	Investigating	Identify a statement supported by information provided in diagrams	C	☒	58
34	Investigating	Identify conditions required for a test to be fair	B	☒	53
35	Interpreting	Describe events portrayed in a diagram using correct terminology	D	B	66
36	Interpreting	Uses information from a diagram to deduce the functions of body parts	B	☒	68
37	Investigating	Identify the steps of an experiment that can be used to achieve its aim	A	☒	47
38	Reasoning/Problem sol	Predict the outcome of an investigation following a change in two variables	D	☒	36
39	Observing/Measuring	Draw a conclusion based on information provided	B	☒	28
40	Reasoning/Problem sol	Determine the amount of water in a cylinder using displacement	B	☒	57